

ASHUTOSH JARAG

jaragashutosh11@gmail.com | +91-9527900909 | <https://github.com/Ashutosh-Jarag>

EDUCATION

Shivaji University, Kolhapur

2022 - 2025

- B.tech in Computer Science(Data Science) | 8.2 CGPA

Dr. D. Y. Patil Polytechnic, Kolhapur

2019 - 2022

- Diploma in Computer Science | 84.23%

EXPERIENCE

CompServ Consultant Pvt Ltd | Intern

Jan. 2025 - Present

- Collaborating with cross-functional teams to design and develop **enterprise-grade Generative AI applications**.
- Built an **LLM-powered automation system** for handwritten bill processing, reducing manual work by **70%**.
- Constructed a **Retrieval-Augmented Generation (RAG)** pipeline to enhance responses from LLMs.
- **Led a team of interns in R&D**, designing project architectures, evaluating **AI frameworks and libraries**, and optimizing system performance.
- Developing a **secure data storage and file tracing platform**, enhancing **data traceability and protection** through encryption and structured audit trails.

Tvarit Solutions Pvt Ltd | Data Science Intern

June 2024 - August 2024

- Collected, cleaned, and prepared large datasets for **Exploratory Data Analysis (EDA)** to support **data-driven decision-making** for senior data scientists.
- **Collaborated with cross-functional teams** to define data requirements, align schemas, and maintain high levels of **data accuracy and consistency**.
- **Refactored procedural Python scripts** into **object-oriented (OOP) structures**, improving **code modularity, readability, and maintainability**.

SKILLS

Programming: Python, Java, R, JavaScript (Basic)

AI & ML: LLMs, NLP, RAG, LangChain, Machine Learning, Deep Learning

Data & Databases: SQL, MySQL, PostgreSQL, NoSQL, VectorDBs

Tools & Frameworks: FastAPI, Flask, PyTorch, TensorFlow, Hugging Face, Docker

Visualization: Power BI, Tableau, Excel, Databricks

Cloud: AWS, GCP

Other Skills: Problem Solving, Leadership, Adaptability, Team Collaboration

PROJECTS

Autolog

April 25

- Developed a **real-time vehicle logging system** using **OCR and LLMs** for automated license plate recognition.
- Reduced manual logging effort by **80%** and improved text extraction accuracy by **35%** through adaptive preprocessing.
- Live Demo - <https://autolog-app.onrender.com/>
- Repo - <https://github.com/Ashutosh-Jarag/AutoLog>

AI Math Agent — AI-Powered Math Tutoring System

Oct 25

- Developed an **AI-driven math assistant** using **FastAPI, React**, and **Google Gemini API** that provides **step-by-step solutions exclusively for math problems**.
- Integrated **Qdrant Vector Database** for semantic retrieval and **MCP (Math Context Processor)** to fetch locally stored **JEE benchmark Q&A** for context-aware responses.
- Enabled **global knowledge retrieval** using **Server.dev API** for web search augmentation when no similar examples exist in the local database.
- Repo - https://github.com/Ashutosh-Jarag/math_agent

DocChat — Retrieval-Augmented Chatbot (RAG)

June 25

- Developed using **LangChain & Gemini API**, to combine document retrieval with generative AI.
- Designed an interactive interface using **Streamlit** that allows users to upload and query PDFs dynamically.
- Implemented **vector-based** document **embedding** for efficient context search and response generation.
- Repo. - <https://github.com/Ashutosh-Jarag/DocChat-AI>

Book Recommendation System

Dec 24

- Built a similarity-based recommendation system using **cosine similarity** and **KNN algorithms** in Python.
- Processed and cleaned user-book datasets to generate meaningful feature matrices.
- Enhanced recommendation precision by combining **popularity metrics** with **user preference-based predictions**.
- Developed a user-friendly interface for displaying top 5 recommended books dynamically.
- Repo. - <https://github.com/Ashutosh-Jarag/Book-Recommender-System>

SMS & Email Spam Detection – NLP Model

Nov 24

- Preprocessed large datasets using **Pandas** and **NLTK** for tokenization, stop-word removal, and feature extraction.
- Trained and compared models using **Logistic Regression**, **Naive Bayes**, and **SVM** to identify the best performer.
- Applied **TF-IDF vectorization** to represent message text for high-dimensional classification.
- Tuned hyperparameters and validated model robustness through **cross-validation** and confusion matrix analysis.
- Repo. - <https://github.com/Ashutosh-Jarag/Sms-Email-spam-Detection>

CERTIFICATIONS

- Data Science & Machine Learning — Udemy
- Advanced SQL — Great Learning
- Data Analytics Essentials — Cisco Networking Academy